ThyroidXL Label-Provenance Audit — Findings and Open Decisions

Date: 24 Oct 2026
Status: Audit complete. No model has been run. Decisions in Section 5 are needed before Analysis Plan v1.1.
Context change: The manuscript was rejected. It is being improved for submission to a different journal. The reviewer comments are kept as an internal quality checklist, not as a revision to answer. Manuscript text changes come last.

1. Files reviewed

File	Role	Main use in the audit
`id2info_eng.json`	Raw per-patient metadata (English), 4,093 patients	Primary source for label provenance
`new_id2info.json`	Raw metadata (Vietnamese), 5,085 patients	Confirmed the meaning of `conclusion` codes; superset of the 4,093
`thyroidxl_clean.csv`	Output of `step1_cleaning.py`, 4,093 rows	Compared case by case with the raw file
`thyroidxl_histonly_multitask.csv`	Earlier histology-only file (image level, 2,225 rows)	Independent cross-check of the histology subset
`README.md`, `data_conversion.ipynb`	Dataset documentation and conversion code	Located the raw metadata; image-label format
`step1_cleaning.py`	Builds the cleaned cohort and the outcome	Origin of the "suspicious" coding
`step4_preop_model.py`	Preoperative models, CV, bootstrap	Leakage review
`sensitivity_suspicious.py`	Former Table S15	Effect of the "suspicious" assumption

2. Findings

2.1 The 94 "suspicious" cases are histopathology-confirmed benign (main finding)

- In the raw data, `conclusion = 4` is `"Benign (Histopathology) - 4"` (English) and `"Lành tính MBH (4)"` (Vietnamese; MBH = mô bệnh học, histopathology).
- 93 of the 94 have a histopathology text: thyroid cyst disease 77, colloid nodule 7, follicular adenoma 5, follicular thyroid tumor 2, Hürthle cell adenoma 1, PTC with chronic thyroiditis 1. One has no text.
- The four codes encode **result × source**:

Code	Meaning	n (of 4,088)
1	Malignant, histopathology	708
2	Malignant, FNAC	521
3	Benign, FNAC	2,765
4	Benign, histopathology	94

- **Origin of the error:** `step1_cleaning.py`, Section B4, comment `# 4 = Suspicious / indeterminate` and `conclusion_map = {..., 4: 'Suspicious'}`. This was an assumption in our code, not a property of the data.
- **Impact on models: none.** The binary outcome is `malignant = conclusion ∈ {1, 2}`, so the 94 were always coded non-malignant, which is correct.
- **Impact on interpretation: substantial.**
  - Former Table S15 ("suspicious cases excluded", `sensitivity_suspicious.py`, `conclusion != 4`) actually removed the 94 best-verified benign cases. Prevalence moved only from 30.1% to 30.8%, which is why results looked unchanged.
  - Every statement describing these cases as suspicious, indeterminate or unresolved is wrong (list in Section 4).
  - The reviewer concern about "94 suspicious cases coded non-malignant" rested on our own mislabelling.

2.2 Label provenance is available per patient

The source of every label is recorded in the raw `Conclusion` string. No reconstruction or assumption is needed.

2.3 Bethesda category × label source (n = 4,088)

Bethesda	Mal-Histo	Mal-FNAC	Ben-FNAC	Ben-Histo	Total
I	2	0	1	3	6
II	15	1	2,755	76	2,847
III	26	0	8	5	39
IV	1	0	0	5	6
V	100	0	1	2	103
VI	425	516	0	0	941

No FNAC	139	4	0	3	146
Total	708	521	2,765	94	4,088

Observations:

- FNAC-based labels are almost entirely Bethesda VI (malignant: 516/521) or Bethesda II (benign: 2,755/2,765).
- 9 FNAC-benign labels have indeterminate cytology (Bethesda III: 8, V: 1).
- 1 FNAC-malignant label has Bethesda II cytology.
- 15 histopathology-malignant cases had Bethesda II cytology (cytology false negatives, consistent with verification by surgery).

2.4 Discordance between label and histopathology text

Patient	Analysis label	Histopathology text	Type
256	Benign (FNAC, Bethesda II)	Papillary thyroid carcinoma	Label error
509	Benign (FNAC, Bethesda II)	Papillary thyroid carcinoma	Label error
370	Benign (Histo, Bethesda III)	PTC, chronic thyroiditis	Label error
366	Malignant (Histo, no FNAC)	Thyroid cyst disease	Label error
1548	Benign (Histo, Bethesda IV)	Follicular thyroid tumor	Uncertain malignant potential
3919	Benign (Histo, Bethesda III)	Follicular thyroid tumor	Uncertain malignant potential

Also:

- 2 FNAC-malignant labels have PTC histopathology (label correct, source understated).
- 1 FNAC-benign label has benign histopathology (label correct, source understated).
- 9 histopathology-coded labels have no histopathology text (8 malignant, all Bethesda VI; 1 benign, Bethesda II).

Clear label errors: 4 of 4,088 (0.1%). Direction: 3 false-negative, 1 false-positive labels.

2.5 Reconciliation of all counts

Count	Source	Explanation
4,093 → 4,088	`step1_cleaning.py`	39 birth-year ages corrected (age = 2022 − year); 5 ages >110 set to missing and excluded (4 FNAC-benign, 1 FNAC-malignant; none with histopathology)
1,229 malignant	Clean file	1,230 in 4,093, minus 1 excluded
798 with histopathology	Raw text present	Matches manuscript
708 histopathology-malignant	`conclusion = 1`	Includes 8 without text and 1 with benign text
796 / 704 / 92	`histonly` file	798 minus 2 follicular thyroid tumors; label derived from histology text. Matches the S-D1 rule exactly

Raw labels were not altered by the pipeline: `conclusion` and histopathology text are identical in the raw and cleaned files for all 4,093 patients.

README is internally inconsistent (train 2,477 + 877 ≠ 3,275; test 386 + 453 ≠ 739; 3,275 + 739 ≠ 4,093). No figure should be taken from the README.

2.6 Unit of analysis

- One nodule per patient: `nodule_2` is empty for all 4,093.
- One row per patient in the cleaned file; 1–10 images per patient (mean 2.8).
- Patient-grouped folds are not needed for the tabular models.

2.7 Subset sizes

Subset	N	Malignant	Benign	Note
Full cohort (Set A)	4,088	1,229	2,859	Mixed reference
S-D1 histopathology only	796	704	92	2 follicular tumors excluded
Set B within S-D1	654	566	88	Below ≥100 benign rule → descriptive only
S-D2 candidate (histo, else Bethesda II/VI)	4,072	1,226	2,846	16 excluded; nearly the full cohort

S-D1 benign spectrum is narrow: 79/92 are thyroid cyst disease; 36 of these 79 had TIRADS 4–5.

2.8 Leakage review of `step4_preop_model.py`

- `StandardScaler` fitted on the training fold only.
- TIRADS baseline uses empirical malignancy rates from the training fold only.

- Hyperparameters fixed in code (no tuning), so nested CV is not required; document as prespecified.
- No imputation; complete-case filtering occurs before CV and is not a fitted step.
- `nodule_area` and `aspect_ratio` are computed row-wise.
- The same `StratifiedKFold` split (fixed seed, same row order) is used for all models and TIRADS, so paired ΔAUC can be computed from existing OOF predictions.
- Feature importance is fitted on the full data, but is descriptive and does not enter any performance metric.

Conclusion: no leakage found. Remaining work for this concern is documentation (pipeline schematic) and adding paired ΔAUC.

2.9 Other observations

- Birth-year correction assumes reference year 2022; the true examination year in ThyroidXL is not recorded in the metadata.
- Set A includes both `age` and `age_55_plus`. Acceptable for prediction, but redundant.
- `new_id2info.json` holds 992 additional patients (and 234 second nodules) with no image lists. They are not part of the released image dataset.

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3. What is not affected

- All primary model inputs and outputs (labels were correct in the binary outcome).
- Set A and Set B AUCs, Brier scores, TIRADS baseline.
- SEER analyses (separate cohort).

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4. Statements that are now incorrect (for the writing stage)

1. Methods: "Suspicious cases (conclusion = 4; n = 94) were retained and coded non-malignant (2,765 benign and 94 suspicious)".
2. Methods: Table S15 described as excluding suspicious cases.
3. Limitations: "some negatives are unresolved rather than confirmed benign (including 12 of the 21 non-malignant cases in Bethesda III–V)".
4. Any text saying histopathology-benign verification is absent; 94 benign cases are histopathology-verified.
5. Supplementary Table S15 title, content and interpretation.
6. Descriptions of the reference standard should list four categories with their sources (Section 2.1 table).

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5. Decisions needed

Study-level (new because of the rejection)

#	Decision	Options	Recommendation
D1	One paper or two	(a) one paper, diagnostic primary; (b) split into ThyroidXL diagnostic paper and SEER nodal paper	Open again now that there is no revision invitation. Splitting gives two focused papers and removes the weakest element (the bridge). Decide before v1.1
D2	Target journal	—	Choose before writing; it sets length, structure and reporting checklist
D3	Plan framing	Update Analysis Plan header and comment map from "revision" to "resubmission with internal checklist"	Update

Diagnostic cohort

#	Decision	Options	Recommendation
D4	Primary outcome rule	(a) dataset labels as released; (b) reference hierarchy: histopathology overrides FNAC where text exists (changes 4 labels)	(b) as primary, (a) as sensitivity. Histopathology-over-cytology is the standard hierarchy, the rule is fixed before any model run, and it affects 0.1% of cases
D5	Former S-D3 / Table S15	(a) remove; (b) keep relabelled as "histopathology-benign excluded"	(a) remove; it has no scientific question now
D6	S-D2	(a) report in one line as near-identical to full cohort; (b) do not construct	(a); S-D1 remains the only stronger-reference analysis
D7	2 follicular thyroid tumors	Keep as non-malignant in primary; exclude from S-D1	Keep as stated
D8	9 FNAC-benign labels with Bethesda III/V	Descriptive only; or exclude in a sensitivity	Descriptive only
D9	9 histopathology-coded cases without text	Keep in primary; outside S-D1 by rule	Keep as stated
D10	992 extra patients	Use or not	Do not use; not part of the released image dataset

	in `new_id2info.json`		
D11	Age reference year 2022	Keep, or check the ThyroidXL paper for collection years	Check the paper; keep 2022 if no year is reported, and state it
D12	`age` and `age_55_plus` both in Set A	Keep or drop the binary flag	Drop `age_55_plus` for new runs (redundant); report as a deviation from v1.0.1

Unchanged by the audit (already in Plan v1)

- S-D1 is a supportive sensitivity analysis; outcome derived from histopathology text for every case.
- Set B within S-D1 is descriptive only (88 benign < 100).
- Bethesda III exploratory; Bethesda IV and V not analysed for discrimination.

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6. SEER audit (Part A of `step1_cleaning.py`, raw file `Seer_original_data.csv`, 61,615 rows)

6.1 Nodal codes 95–97 treated as counts (main finding)

`step1_cleaning.py` keeps any value <98 as a count. In SEER, examined 95/96/97 and positive 95/97 are special codes (aspiration only; sampling or dissection with unknown number; positive nodes with unknown number). LNR and LODDS were therefore computed from pseudo-counts for these records.

Group	Valid by old rule	CSS events	With codes 95–97	CSS events	Clean	CSS events
All histologies	33,488	576	817 (2.4%)	143 (25%)	32,671	433
DTC (corrected map)	32,146	313	656	55 (18%)	31,490	258
ATC (8020/8021)	298	179	84	62	214	117

The affected records are few but carry a quarter of nodal-cohort CSS deaths, because aspiration-only or unknown-number records are concentrated in advanced disease. All previous LNR/LODDS results include them.

Count update (24 Oct 2026): the figures above (817 / 656) required the other nodal field to be <98. The final definition counts any record with examined 95/96/97 or positive 95/97 in the full cleaned file: **840 records (145 CSS events), of which DTC 675 (56)**. These final counts are used in the Analysis Plan, Stage-2 report and verification script.

Other nodal checks: no record with positive > examined once special codes are removed; 294 records have examined = 90 (“90 or more”), used as 90.

6.2 ICD-O-3 histology map

Code(s)	n	CSS	Current map	Issue
8050 papillary carcinoma NOS	284	6	Other	Should be PTC
8331/8332 follicular, well/moderately differentiated	28	2	Other	Should be FTC
8350 nonencapsulated sclerosing carcinoma	111	2	FTC	Papillary family; should be PTC
8022, 8030–8035 anaplastic variants	13	7	Other	Should be ATC
8347 mixed medullary–papillary	43	—	PTC	Mixed medullary; should not be DTC
8290 oncocytic (Hürthle) carcinoma	1,305	25	FTC	Acceptable in DTC; state explicitly
8000/8001/8010 malignant neoplasm or carcinoma NOS	835	246	Other	Unknown histology, high mortality; outside DTC

Effect on DTC is small in size (≈+300 patients, ≈+8 events), but the DTC definition must use the corrected map.

6.3 Other SEER observations

- 118 records are “Dead (missing/unknown COD)” and are coded as non-events (alive) for CSS.
- 141 records have missing survival time (all deceased; known limitation).
- Corrected DTC nodal cohort: ≈31,490 patients and ≈258 CSS events before survival-time and covariate filtering, lower than the ≈310 assumed in Plan v1.

6.4 SEER decisions needed

#	Decision	Recommendation
S1	Records with nodal codes 95–97	Exclude from all LNR/LODDS/N models (ratios not computable); describe them as a separate group and state the selection consequence
S2	Examined = 90 (“90 or more”)	Keep as 90; note
S3	Histology map	Apply the corrections in 6.2; DTC = PTC (8050, 8260, 8337, 8339–8344, 8350) + FTC (8290, 8330, 8331, 8332, 8335)
S4	Unknown cause of	Exclude from CSS analyses (primary); sensitivity counting them as CSS events

	death (118)	
S5	Events per parameter	≈258 events is close to the ≥20 EPP rule. Prespecify T collapsed to T1/T2/T3/T4 (3 df) before fitting, so the largest model has ≈10 parameters; ridge fallback rule unchanged